

Exercise I. (course)

[7 marks]

(4)+(3) Let $f : [a, b] \mapsto \mathbb{R}$ be a bounded function. Prove that f is Riemann integrable on $[a, b]$ if and only if for every $\epsilon > 0$, there exists a partition P_ϵ such that $U(f; P_\epsilon) - L(f; P_\epsilon) < \epsilon$.

Exercise II.

[8 marks]

Let $g(x) = \int_0^{\sin^2 x} \arcsin \sqrt{t} dt + \int_0^{\cos^2 x} \arccos \sqrt{t} dt$.

(2+2+1/2) 1. Calculate $g'(x)$ and show that the domain of differentiability of g is $\mathbb{R}$.

(1) 2. Show that $g'(x) = 0$ on $[0, \frac{\pi}{2}]$.

(1) 3. Compute $g(\frac{\pi}{4})$.

Hint : You may use $\arcsin x + \arccos x = \frac{\pi}{2} \quad \forall x \in [-1, 1]$.

(1/2) 4. Deduce the values of $\int_0^1 \arcsin \sqrt{t} dt$ and $\int_0^1 \arccos \sqrt{t} dt$.

Exercise III.

[9 marks]

Compute the following integrals :

(1/2) (1) $\int \frac{x}{\cos^2 x} dx,$

(1/2) (2) $\int \frac{1}{-x^2 + x + 1} dx,$

(1/2) (3) $\int \frac{1}{x^2 + x + 5} dx,$

(1/2) (4) $\int \frac{x-3}{x^2 + x + 5} dx,$

(1/2) (5) $\int \frac{\tan^2 x}{\cos^2 x + 2 \sin^2 x} dx,$

(1/2) (6) $\int_{-1}^1 \frac{(\sin x) e^{x^2 \cos x}}{\ln(x^2 + 2)} dx.$

Exercise IV.

[3 marks]

Give the general form, without calculating the coefficients, of the partial fraction decomposition for the following rational function

(1/2 x 6)

$$f(x) = \frac{x+1}{x^2(x+1)^2(x^2+x+1)^2}.$$

Exercise V.

[3 marks]

(2+1) Let $n \in \mathbb{N}$ and $I_n = \int x^n e^{-x^3} dx$. Find a relation between I_{n+3} and I_n .

Exercise I Let $f: [a, b] \rightarrow \mathbb{R}$ be a bdd fn.

$\Rightarrow$ Given; $f \in R([a, b]) \Leftrightarrow U(f) = L(f)$

claim; $\forall \epsilon > 0, \exists P_\epsilon$ partition of $[a, b]$ s.t. $U(f; P_\epsilon) - L(f; P_\epsilon) < \epsilon$

Let $\epsilon > 0$, then by def of sup & inf, $\exists$ two partitions P_1 & P_2 of $[a, b]$ s.t.
 $U(f; P_1) < U(f) + \frac{\epsilon}{2}$ and $L(f; P_2) > L(f) - \frac{\epsilon}{2}$. (1 pt)

Let $P_\epsilon = P_1 \cup P_2$ be the common refinement & since $L(f) = U(f)$,
 $U(f; P_\epsilon) - L(f; P_\epsilon) \leq U(f; P_1) - L(f; P_2)$ (1 pt)
 $\leq [U(f; P_1) - U(f)] + [L(f) - L(f; P_2)] < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon$ (1 pt)

$\Leftarrow$ Given; $\forall \epsilon > 0, \exists P_\epsilon$ part of $[a, b]$ s.t. $U(f; P_\epsilon) - L(f; P_\epsilon) < \epsilon$.
 claim; $f \in R([a, b]) \Leftrightarrow U(f) = L(f) \Leftrightarrow \forall \epsilon > 0, \underbrace{U(f) - L(f)}_{\text{positive}} < \epsilon$. (1 pt)

Let $\epsilon > 0, \exists P_\epsilon$ part of $[a, b]$ s.t. $U(f; P_\epsilon) - L(f; P_\epsilon) < \epsilon$.
 Then $U(f) - L(f) \leq U(f; P_\epsilon) - L(f; P_\epsilon) < \epsilon$. It follows that
 $U(f) - L(f) = 0$ and so $f \in R([a, b])$. (2 pt)

Exercise II

$$\text{Let } g(x) = \int_0^{\sin^2 x} \arcsin \sqrt{t} \, dt + \int_0^{\cos^2 x} \arccos \sqrt{t} \, dt.$$

$$\text{1. * } g'(x) = ? : g'(x) = (\sin^2 x)' \arcsin \sqrt{\sin^2 x} + (\cos^2 x)' \arccos \sqrt{\cos^2 x} \quad (1 \text{ pt})$$

$$= 2 \sin x \cos x \arcsin |\sin x| - 2 \sin x \cos x \arccos |\cos x|$$

$$= \sin 2x [\arcsin |\sin x| - \arccos |\cos x|] \quad (1 \text{ pt})$$

* Domain of diff of g :

for the fn, defined by $x \mapsto \int_0^{\sin^2 x} \arcsin \sqrt{t} \, dt$; let $f(t) = \arcsin \sqrt{t}$

(1/2 pt) ① f is cont on $D = [0, 1]$

(1/4 pt) ② $u(x) = \sin x$ & $v(x) = 0$ are diff on $\mathbb{R} = I_1$

(1/4 pt) ③ $u(x) \in [0, 1]$ & $v(x) \in [0, 1] \Leftrightarrow x \in \mathbb{R} = I_2$ } $\Rightarrow I_1 = I_1 \cap I_2 = \mathbb{R}$ domain of diff (1/2 pt)

Similarly for $x \mapsto \int_0^{\cos^2 x} \arccos \sqrt{t} \, dt$, domain of diff is $J_2 = \mathbb{R}$.

Thus Domain of diff of g is $J_1 \cap J_2 = \mathbb{R}$.

(1/2 pt)

2. $g'(x) = 0$ on $[0, \pi/2]$.

Let $x \in [0, \pi/2]$, $|\sin x| = \sin x$ & $|\cos x| = \cos x$. (1 pt)

So $g'(x) = \sin 2x [\arcsin \sin x - \arccos \cos x] = \sin 2x [x - x] = 0$.

3. $g(\pi/4) = ?$ $g(\pi/4) = \int_0^{\pi/4} \arcsin \sqrt{t} dt + \int_0^{\pi/4} \arccos \sqrt{t} dt$ (1 pt)
 $= \int_0^{\pi/4} (\arcsin \sqrt{t} + \arccos \sqrt{t}) dt$
 $= \int_0^{\pi/4} \pi/4 dt = \frac{\pi}{4}$.

4. values of $\int_0^1 \arcsin \sqrt{t} dt$ & $\int_0^1 \arccos \sqrt{t} dt$?

Since $g'(x) = 0$ on $[0, \pi/2]$, then $g(x) = \text{const} = g(\pi/4) = \frac{\pi}{4}$. (1/2 pt)

Hence $\int_0^{\sin^2 x} \arcsin \sqrt{t} dt + \int_0^{\cos^2 x} \arccos \sqrt{t} dt = \frac{\pi}{4} \quad \forall x \in [0, \pi/2]$.

(1/2 pt) for $x=0$, we obtain $\int_0^1 \arccos \sqrt{t} dt = \pi/4$ and
 (1/2 pt) for $x=\pi/2$, we obtain $\int_0^1 \arcsin \sqrt{t} dt = \pi/4$.

Exercise III Compute the following integrals:

(1) $\int \frac{x \cdot \frac{1}{\cos^2 x}}{\frac{1}{\cos^2 x}} dx = x \cdot \tan x - \int \tan x dx$ (1/2 pt)
 $= x \tan x - \int \frac{\sin x}{\cos x} dx$
 $= x \tan x + \int \frac{d(\cos x)}{\cos x}$
 $= x \tan x + \ln |\cos x| + C$ (1/2 pt)

$u = x \Rightarrow u' = 1$
 $v' = \frac{1}{\cos^2 x} \Rightarrow v = \tan x$

(2) $\int \frac{1}{-x^2+x+1} dx = \int \frac{1}{-x^2-x-1} dx = \int \frac{1}{-x^2-x+\frac{1}{4}-\frac{5}{4}} dx$

(1 pt) $= - \int \frac{d(x-1/2)}{(x-1/2)^2 - (\sqrt{5}/2)^2} = - \frac{1}{\sqrt{5}} \ln \left| \frac{(x-1/2) - \sqrt{5}/2}{(x-1/2) + \sqrt{5}/2} \right| + C$ (1/2 pt)
 $= - \frac{1}{\sqrt{5}} \ln \left| \frac{x - \frac{1+\sqrt{5}}{2}}{x + \frac{\sqrt{5}-1}{2}} \right| + C$

(3) $\int \frac{1}{x^2+x+5} dx = \int \frac{1}{x^2+x+\frac{1}{4}+\frac{19}{4}} dx = \int \frac{d(x+1/2)}{(x+1/2)^2 + \frac{19}{4}}$ (1 pt)
 $= \frac{1}{\sqrt{19/4}} \arctan \frac{x+1/2}{\sqrt{19/4}} + C$

$$\begin{aligned}
 (4) \int \frac{x-3}{x^2+x+5} dx &= \frac{1}{2} \int \frac{2x+1}{x^2+x+5} dx + \int \frac{-1/2-3}{x^2+x+5} dx \\
 &= \frac{1}{2} \int \frac{d(x^2+x+5)}{x^2+x+5} + \frac{7}{2} \int \frac{dx}{x^2+x+5} \quad (1/2 \text{ pt}) \\
 &\stackrel{\text{using (3)}}{=} \frac{1}{2} \ln|x^2+x+5| - \frac{7}{2} \cdot \frac{1}{\sqrt{19/4}} \arctan \frac{x+1/2}{\sqrt{19/4}} + \text{const.} \quad (1/2 \text{ pt})
 \end{aligned}$$

$$\begin{aligned}
 (5) \int \frac{\tan^2 x}{\cos^2 x + 2 \sin^2 x} dx &\quad \text{B.R. rule: } w(u+n) = w(x) \\
 &\quad ; t = \tan x \Rightarrow dt = \frac{1}{\cos^2 x} dx = (1 + \tan^2 x) dx \\
 &= \int \frac{\sin^2 x}{\cos^2 x + 2 \sin^2 x} \frac{dx}{\cos^2 x} = \int \frac{1}{\frac{\cos^2 x}{\sin^2 x} + 2} \frac{dx}{\cos^2 x} \\
 &= \int \frac{1}{\frac{1}{\tan^2 x} + 2} \frac{dx}{\cos^2 x} = \int \frac{1}{\frac{1}{t^2} + 2} dt = \int \frac{t^2 + 2^{-2}}{2 + t^2} dt \quad (1/2 \text{ pt})
 \end{aligned}$$

$$= \int \frac{dt}{t^2 + 2} - 2 \int \frac{dt}{t^2 + 2} = t - 2 \cdot \frac{1}{\sqrt{2}} \arctan \frac{t}{\sqrt{2}}$$

$$= \tan x - \frac{2}{\sqrt{2}} \arctan \frac{\tan x}{\sqrt{2}} + \text{const.} \quad (1/2 \text{ pt})$$

$$(6) \int_{-1}^1 \frac{(\sin x) e^{x^2 \cos x}}{\ln(x^2+2)} dx = 0 \quad \because \text{the integrand is odd \& cont on } [-1,1]. \quad (1/2 \text{ pt})$$

Exercise IV

$$\begin{aligned}
 f(x) &= \frac{x+1}{x^2(x+1)^2(x^2+x+1)^2} \\
 &\stackrel{(1/2 \text{ pt})}{=} \frac{1}{\underbrace{(x+1)}_{\text{mon rep}} \underbrace{x^2}_{\text{or}} \underbrace{(x^2+x+1)^2}_{\text{quadratic \& no}}} = \frac{a}{(x+1)} + \frac{b}{x} + \frac{c}{x^2} + \frac{dx+e}{x^2+x+1} + \frac{kx+l}{(x^2+x+1)^2} \quad (1/2 \text{ pt})
 \end{aligned}$$

Exercise V

$$\begin{aligned}
 I_n &= \int \frac{x^n}{u'} \frac{e^{-x^3}}{v} dx = \frac{x^{n+1}}{n+1} e^{-x^3} - \int \frac{x^{n+1}}{n+1} (-3x^2) e^{-x^3} dx \quad (2 \text{ pts}) \\
 &= \frac{x^{n+1}}{n+1} e^{-x^3} + \frac{3}{n+1} \int x^{n+3} e^{-x^3} dx \\
 &= \frac{x^{n+1}}{n+1} e^{-x^3} + \frac{3}{n+1} I_{n+3} \quad (1 \text{ pt})
 \end{aligned}$$